

3.3 Review: Linear Independence and Basis

Proposition. (3.4) Let $A \in \mathcal{M}_{nn}$. Then A is nonsingular if and only if its column vectors form a basis for $\mathbb{R}^n$.

Example. Show that $\mathbf{v}_1 = (1, 0, 3)$, $\mathbf{v}_2 = (1, 1, 2)$, $\mathbf{v}_3 = (1, 1, 1)$, and $\mathbf{b} = (3, 0, 1)$. Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbb{R}^3$.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So it is a basis by the proposition.

- $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\Rightarrow$ the vectors are linearly independent.
- $A\mathbf{x} = \mathbf{b}$ has a unique solution $\Rightarrow$ any vector in $\mathbb{R}^3$ is a linear combination of the vectors.
- Together this means the vectors form a basis of $\mathbb{R}^3$.

To write $\mathbf{b}$ as a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, reduce

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 3 & 2 & 1 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 8 \end{array} \right].$$

Which gives the *coordinates* of $\mathbf{b}$ with respect to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$,

$$3 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} - 8 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + 8 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

3.4 Dimensions and Its Consequences.

- There are many bases for a given subspace $V \subset \mathbb{R}^n$, but they ALL have the same number of elements.

Proposition. Let $V \subset \mathbb{R}^n$ be a subspace, let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a basis for V , and let $\{\mathbf{w}_1, \dots, \mathbf{w}_\ell\} \in V$. If $\ell > k$ then $\{\mathbf{w}_1, \dots, \mathbf{w}_\ell\}$ must be linearly dependent.

Example. Let $V = \{\mathbf{v}_1, \dots, \mathbf{v}_{n+1}\}$ for $v_i \in \mathbb{R}^n$.

1. Is V a basis for $\mathbb{R}^n$?
2. Is V linearly independent or dependent?

Theorem. Let $V \subset \mathbb{R}^n$ be a subspace, and let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_\ell\}$ be two bases for V , then $k = \ell$.

- Recall that the standard basis for $\mathbb{R}^n$ has n vectors so any other basis of $\mathbb{R}^n$ will also have n vectors.

Definition. The *dimension* is a subspace $V \subset \mathbb{R}^n$ is the number of vectors in any basis for V , denoted $\dim V$.

- By convention $\dim\{\mathbf{0}\} = 0$.
- $\dim \mathbb{R}^n = n$ ($\mathbb{R}^n$ is a subspace of itself).

Example. (HW 1.b) Find a basis for each of the given subspaces and determine its dimension.

1. $V = \{\mathbf{x} \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0, x_2 + x_4 = 0\} \subset \mathbb{R}^4$.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right] \Rightarrow x_1 = -x_3 \text{ and } x_2 = -x_4$$

$\Rightarrow$ the general solution is $\mathbf{x} = x_3(-1, 0, 1, 0) + x_4(0, -1, 0, 1)$. So V is spanned by $(-1, 0, 1, 0)$ and $(0, -1, 0, 1)$. It's easily checked that these vectors are linearly independent. So V is a basis with $\dim V = 2$.

2. $V = \text{span} \{(3, -2, 4), (-3, 2, -4), (-6, 4, -8)\}$.

$$\begin{bmatrix} 3 & -3 & -6 \\ -2 & 2 & 4 \\ 4 & -4 & -8 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

So $(3, -2, 4)$ (as is the other vectors) is a basis and $\dim V = 1$.